

CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ SOLUTIONS

Chapter 12: Surface Areas and Volumes

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Surface area

1. A solid sphere is cut into two hemispheres. Find the ratio of the surface area of the sphere to the total surface area of the two hemispheres together. (a) 1 : 1 (b) 1 : 4 (c) 2 : 3 (d) 3 : 2

CBSE 2024, 1 mark

SOLUTION

Surface area of the sphere = $4\pi r^2$

Each solid hemisphere has total surface area $3\pi r^2$, so the two together have $6\pi r^2$.

Ratio = $4\pi r^2 : 6\pi r^2 = 2 : 3$

Answer: (c)

2 Volume

2. The largest possible right circular cone is cut from a solid cube of edge 2 cm. Find its volume. (a) $\frac{4\pi}{3}$ cu cm (b) $\frac{5\pi}{3}$ cu cm (c) $\frac{8\pi}{3}$ cu cm (d) $\frac{2\pi}{3}$ cu cm

CBSE 2024, 1 mark

SOLUTION

The largest cone has base diameter 2 cm and height 2 cm, so $r = 1$ cm and $h = 2$ cm.

Volume = $\frac{1}{3}\pi r^2 h = \frac{1}{3} \times \pi \times 1 \times 2 = \frac{2\pi}{3}$ cu cm

Answer: (d)

3. A hollow cylinder is 14 cm long. Its outer radius is 1 cm more than its inner radius, and the metal in it has volume 176 cm³. Find the two radii.

CBSE 2024, 3 marks

SOLUTION

Let the outer and inner radii be R and r . Then $R - r = 1$.

Volume of metal = $\pi(R^2 - r^2)h = \frac{22}{7} \times (R + r)(R - r) \times 14 = 176$

$44(R + r) = 176$, so $R + r = 4$.

Solving with $R - r = 1$: $R = 2.5$ and $r = 1.5$.

Answer: outer radius 2.5 cm, inner radius 1.5 cm

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